

# Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity

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## Abstract

Land-use conflicts limit conventional agrivoltaics due to solar competition. Here, we present a modelled, quantum-guided agrivoltaic system incorporating a semi-transparent organic photovoltaic layer coupled with a plasmonic nanoparticle-on-mirror (NPoM) cavity that acts as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of 19.6 kg (CO<sub>2</sub>)e/m<sup>2</sup>/yr. Secure IoT telemetry is maintained over a BB84 quantum link. Finally, cooperative economic modeling for a Cameroonian micro-module demonstrates a payback period of 4.23 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m<sup>2</sup> cooperative case.

**Keywords:** quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration

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<sup>1</sup> Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cul-  
<sup>2</sup> tivation to mitigate the land-use competition between food production and  
<sup>3</sup> renewable energy generation [1, 2]. Land-use optimisation and mobile-PV  
<sup>4</sup> foundations for this field were established in *Applied Energy* [3, 4], and recent

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typology reviews systematize overhead, vertical, and tracking configurations with land-equivalent-ratio and levelized-cost metrics [5]. However, conventional semi-transparent PV claddings act as static optical filters that reduce irradiance uniformly, trading electrical output for crop biomass without optimization of the spectral light-sharing window [6, 7]. The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [8]. This spectral specificity allows the PV layer to transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a spectral co-design problem.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as a model system for exploring this concept [9]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [10] and room temperature [11]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — where electronic transitions hybridize with underdamped protein vibrations — plays a central role in directing energy flow [12, 13]. Non-Markovian frameworks such as the hierarchical equations of motion [14] and the process tensor hierarchy of pure states (PT-HOPS) [15, 16] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare — the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic product [17] — and photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [18, 19].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [20]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. A related study from our group, published in the *Journal of Physical Chemistry Letters*, reports the filter, excitation-transport-rate, and single-pass-conversion-efficiency analysis [21]; the present work adds the monolithic photothermal-heating hardware, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting

complexes under ambient conditions [22], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [23]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield  $\Phi_{\text{FT}}$  as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [24] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a  $9 \times 9$  dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic quantum optimization onto a macroscopic multi-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [25] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [26] for secure IoT telemetry. This multi-scale perspective [27] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modelling framework for quantum-guided agrivoltaics (Fig. 4).

The central question guiding this work is falsifiable: can exact non-Markovian quantum dynamics (PT-HOPS) jointly optimize excitation-transport efficiency and power-conversion efficiency in a sensing-integrated photothermal agrivoltaic stack — and does the multifunctional net energy benefit (heat + power + crop + sensing, in  $\$/\text{W} + \$/\text{kg} + \$/\text{m}^3$ ) stay positive under field constraints? We answer it affirmatively but conditionally, within the modelled  $500 \text{ m}^2$  cooperative case detailed below.

## Results

### *8-Site FMO hamiltonian with reaction center trap*

We construct an 8-site model of the FMO complex, extending the standard 7-site framework [28] by including a coupling to the eighth bacteriochlorophyll that bridges the baseplate to the reaction center (Site 8,  $\sim 200$

Table 1: **8-site FMO site energies and key electronic couplings** ( $\text{cm}^{-1}$ ). Full coupling matrix (28 unique pairs) provided in [Section S1](#).

| Parameter       | Site 1 | Site 2 | Site 3 | Site 4 | Site 5 | Site 6 | Site 7 | Site 8 |
|-----------------|--------|--------|--------|--------|--------|--------|--------|--------|
| $\varepsilon_n$ | 280    | 420    | 0      | 110    | 270    | 500    | 310    | 200    |
| $J_{n,n+1}$     | —      | -87.7  | 30.8   | -53.5  | -70.7  | 81.1   | 39.7   | 12.0   |

78  $\text{cm}^{-1}$  above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

79 where  $\varepsilon_n$  are the site energies (relative to Site 3),  $J_{nm}$  are the inter-site  
80 electronic couplings (all 28 unique pairs defined in [Section S1](#)), and  $\Gamma_{\text{RC}} =$   
81  $0.15 \text{ ps}^{-1}$  is the irreversible reaction center trapping rate [29]. The anti-  
82 Hermitian term  $-i\Gamma_{\text{RC}}$  on Sites 3 and 4 (physical indices; Python indices 2  
83 and 3) models the irreversible charge separation at the reaction center. All 28  
84 inter-site couplings are validated to be finite, and the Hamiltonian is verified  
85 to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

#### 86 *NPoM plasmonic strong coupling*

87 The OPV-FMO interface is mediated by a nanoparticle-on-mirror  
88 (NPoM) architecture [24], where a gold nanoparticle (diameter 80 nm) is  
89 separated from a gold mirror by a 1 nm graphene monolayer spacer. The  
90 graphene barrier prevents Dexter-type electron transfer while maintaining  
91 electromagnetic field enhancement. The NPoM cavity supports an intense  
92 localized surface plasmon mode that couples coherently to both the OPV  
93 exciton band and the FMO optical transitions.

94 The single-emitter coupling strength follows the cavity quantum electro-  
95 dynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

96 with  $g_{\text{ref}} = 120 \text{ cm}^{-1}$  at  $V_{\text{ref}} = 1 \text{ nm}^3$  [24]. At the sub-nanometer mode vol-  
97 umes accessible in NPoM cavities ( $V_{\text{mode}} < 1 \text{ nm}^3$ ),  $g_0$  ranges from  $120 \text{ cm}^{-1}$   
98 to over  $300 \text{ cm}^{-1}$ , and the collective coupling  $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$  places the sys-  
99 tem in the strong-coupling regime ( $g_{\text{eff}} > \kappa$ , where  $\kappa \approx 800 \text{ cm}^{-1}$  is the plas-  
100 mon linewidth). A numerical guardrail caps  $g_0$  at  $1000 \text{ cm}^{-1}$  when  $V_{\text{mode}} \rightarrow 0$

101 to prevent divergences. Recent advances in quantum-tunnel-junction plas-  
 102 monic sources have demonstrated that such sub-nanometer cavities can be  
 103 self-illuminating, eliminating the need for bulky external optics [30].

104 The dressed Hamiltonian expands to  $9 \times 9$ :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

105 where the plasmon mode at frequency  $\omega_{\text{pl}} \approx 12\,500\text{ cm}^{-1}$  couples to the  
 106 primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength  
 107  $g_0$ . Figure 1a shows the energy level diagram.

### 108 *Non-Markovian quantum dynamics*

109 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-  
 110 ism with production parameters: hierarchy depth  $L = 8$ ,  $K = 2$  Matsubara  
 111 terms, time step  $\Delta t = 0.2\text{ fs}$ , and  $n = 100$  disorder realizations ( $\sigma = 50\text{ cm}^{-1}$   
 112 static disorder) [15, 16]. The bath is modeled as a Drude–Lorentz over-  
 113 damped component ( $\lambda_D = 35\text{ cm}^{-1}$ ,  $\gamma_D = 50\text{ cm}^{-1}$ ) plus 12 underdamped  
 114 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-  
 115 gence is verified through hierarchy depth sweeps ( $L = 6, 7, 8$ ) and Matsubara  
 116 truncation tests ( $K = 1, 2, 3$ ); the maximum relative error between  $L = 7$   
 117 and  $L = 8$  populations is  $3.1 \times 10^{-11}$ , and between  $K = 2$  and  $K = 3$  is  
 118  $3.3 \times 10^{-5}$  (Section S2). The computational feasibility, scaling, and memory-  
 119 aware scheduling for these high-order simulations are detailed and validated  
 120 in our companion study [20].

121 Figure 1b shows the exciton population dynamics under broadband *vs.*  
 122 dual-band filtered excitation. The filtered case (solid lines) maintains higher  
 123 population on the trapping sites (Sites 3 and 4), leading to enhanced forward  
 124 transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

125 where  $P_j(t) = \rho_{jj}(t)$  are the site populations and  $t_{\text{max}} = 1\text{ ps}$ .

126 Under dual-band filtering,  $\Phi_{\text{FT}} = 0.89(3)$ , compared to  $\Phi_{\text{FT}} = 0.71(2)$  for  
 127 broadband excitation — a relative enhancement of 25% (Figure 1c). These  
 128 values are consistent with the spectral coherence protection mechanism es-  
 129 tablished for the 7-site FMO complex [20]. The coherence lifetime  $\tau_c$ , defined

as the  $1/e$  decay time of the  $l_1$ -norm coherence  $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$  [31], extends from  $280 \pm 25$  fs (broadband) to  $420 \pm 35$  fs (filtered) — a 50 % increase that is causally linked to the yield gain: the extended coherence suppresses the pure-dephasing rate, lengthening the effective transport mean-free path and thereby increasing the fraction of the excitation wavepacket that reaches the reaction center before decohering. The mechanism is identical to that established for the 7-site system [20]: the dual-band filter selectively populates polaron-dressed states whose residual reorganization energy  $\lambda_{\text{res}}$  is reduced by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

### NPoM cavity volume dependence of exciton transport

The nanoparticle-on-mirror cavity couples the FMO complex to the OPV active layer through the strong coherent interaction described above, but the plasmonic mode also acts as an additional population decay channel that competes with the reaction center trapping. To quantify this trade-off, we perform a sensitivity scan of the NPoM mode volume  $V_{\text{mode}}$ , which controls the single-emitter coupling strength via  $g_0 \propto V_{\text{mode}}^{-1/2}$  (Equation (2)). Smaller mode volumes produce stronger plasmon–exciton coupling but also enhance the plasmon’s role as a population sink, diverting excitation away from the reaction center.

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM):  $\Phi_{\text{FT}} = 0.98$ . All NPoM runs with  $L = 8$ ,  $K = 2$ ,  $\Delta t = 0.2$  fs,  $t_{\text{max}} = 1$  ps. The  $V_{\text{mode}} = 1.2 \text{ nm}^3$  entry is validated with  $n = 20$  trajectories ( $\Phi_{\text{FT}} = 0.0799 \pm 0.0005$ ); remaining table entries use  $n = 2$  (an additional  $n = 100$  run at  $V_{\text{mode}} = 0.8 \text{ nm}^3$  is discussed in the text). SERS enhancement  $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$  via Purcell factor scaling, normalized to  $\text{EF}_{\text{SERS}} = 10^2$  at  $V_{\text{mode}} = 0.8 \text{ nm}^3$  [24].

| $V_{\text{mode}} \text{ (nm}^3\text{)}$ | $\Phi_{\text{FT}}$ | $\Delta$ vs. baseline (%) | $\text{EF}_{\text{SERS}}$ |
|-----------------------------------------|--------------------|---------------------------|---------------------------|
| 0.2                                     | 0.050 5            | −94.8                     | 1 600                     |
| 0.4                                     | 0.060 5            | −93.8                     | 400                       |
| 0.6                                     | 0.072 7            | −92.6                     | 180                       |
| 0.8                                     | 0.076 8            | −92.2                     | 100                       |
| 1.0                                     | 0.079 5            | −91.9                     | 64                        |
| 1.2                                     | 0.079 9            | −91.8                     | 44                        |
| 1.4                                     | 0.079 7            | −91.9                     | 33                        |

Table 2 summarizes the forward transfer yield across seven mode volumes spanning the experimentally accessible sub-nanometer range [24]. The yield

151 increases monotonically with mode volume, from  $\Phi_{\text{FT}} = 0.0505$  at  $V_{\text{mode}} =$   
152  $0.2 \text{ nm}^3$  to a maximum of  $\Phi_{\text{FT}} = 0.0799$  at  $V_{\text{mode}} = 1.2 \text{ nm}^3$ , with a slight  
153 decrease at  $1.4 \text{ nm}^3$ . The trend is logarithmic with diminishing returns above  
154  $1.0 \text{ nm}^3$ ; increasing the volume beyond  $1.0 \text{ nm}^3$  recovers only 1 % of the  
155 baseline yield.

156 The physical origin of this suppression is the strong-coupling-induced hy-  
157 bridization of excitonic and plasmonic states. The dressed eigenstates acquire  
158 a finite plasmonic character, which decays via the fast plasmon relaxation  
159 pathway ( $\kappa \approx 800 \text{ cm}^{-1}$ ,  $Q \approx 15$ ) rather than populating the reaction center.  
160 At  $V_{\text{mode}} = 0.2 \text{ nm}^3$ , where  $g_0$  is largest, the plasmonic admixture in the  
161 lower polariton branch reaches  $\sim 15 \%$ , diverting a corresponding fraction of  
162 the excitation flux. As  $V_{\text{mode}}$  increases,  $g_0$  decreases, the polariton splitting  
163 narrows, and the excitonic character of the bright states is partially restored.

164 Despite the monotonic recovery, all NPoM-on yields remain suppressed  
165 by more than 90 % relative to the pristine FMO baseline ( $\Phi_{\text{FT}} = 0.98$ ). This  
166 indicates that, in the strong collective coupling regime accessible to NPoM  
167 cavities ( $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ ), the plasmonic decay pathway fundamentally  
168 limits the maximum achievable transport efficiency. The SERS enhance-  
169 ment factor ( $\sim 10^2$  at  $V_{\text{mode}} = 0.8 \text{ nm}^3$ ) scales as  $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ ,  
170 following the Purcell enhancement of the local field at both the excitation  
171 and emission wavelengths. Table 2 includes the calculated SERS enhance-  
172 ment for each mode volume, revealing a four-order-of-magnitude range ( $33$  to  
173  $1.6 \times 10^3$ ) across the scanned volumes. The optimal volume represents a com-  
174 promise:  $V_{\text{mode}} \approx 1.2 \text{ nm}^3$  maximizes transport yield ( $\Phi_{\text{FT}} = 0.0799$ ) while  
175 retaining SERS sensitivity ( $\text{EF}_{\text{SERS}} \approx 44$ ), sufficient for detecting  $180 \text{ cm}^{-1}$   
176 Franck–Condon mode populations at the single-FMO-complex level [32]. By  
177 contrast,  $V_{\text{mode}} = 0.2 \text{ nm}^3$  maximizes SERS sensitivity ( $1.6 \times 10^3$ ) at the cost  
178 of the lowest transport yield ( $\Phi_{\text{FT}} = 0.0505$ ), a trade-off better suited to  
179 diagnostic-only applications where photochemical efficiency is secondary. At  
180 the reference volume, a dedicated  $n = 100$  production run at  $V_{\text{mode}} = 0.8 \text{ nm}^3$   
181 confirms the same limiting behaviour ( $\Phi_{\text{FT}} = 0.0768$ , final plasmon popula-  
182 tion 0.79; Section S11).

183 Second, reducing the temperature to 77 K (analyzed as a theoretical limit-  
184 case for thermal phonon suppression in Section S11.2) doubles the trapping  
185 yield to  $\Phi_{\text{FT}} = 0.1685$ . In the main agrivoltaic operating regime at 295 K,  
186 the maximum trapped population on sites 3+4 remains at  $3.8 \times 10^{-3}$  for  
187  $V_{\text{mode}} = 1.2 \text{ nm}^3$ , requiring the multi-scale sentinel fraction strategy to bypass  
188 plasmonic loading.

189 All room-temperature runs exhibit perfect trace conservation ( $\text{Tr } \rho =$   
190 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.  
191 The  $V = 0.8 \text{ nm}^3$  configuration was run with  $n = 100$  disorder realizations;  
192 the  $V = 1.2 \text{ nm}^3$  configuration was validated with  $n = 20$  trajectories ( $\Phi_{\text{FT}} =$   
193 0.0799, within 0.6 % of the  $n = 2$  estimate of 0.0804). The baseline NPoM-off  
194 run ( $n = 20$ ) is shown as a limiting-case reference; quantitative comparison  
195 across configurations is limited by the differing trajectory counts.

### 196 *Global canopy yield integration*

197 A key insight of our multi-scale framework is that the NPoM SERS diag-  
198 nostic mode (which suppresses local excitonic transport due to plasmon pop-  
199 ulation loading) need only cover a 1 % sentinel fraction  $\alpha$  of the canopy area.  
200 The remaining 99 % operates in passive OPV mode, preserving the near-  
201 baseline forward transfer yield  $\Phi_{\text{FT}}^{\text{passive}} = 0.980$ . The area-weighted *global*  
202 canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

203 yielding  $\Phi_{\text{FT}}^{\text{global}} = 0.971$  at  $\alpha = 0.01$  and  $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$  (or 0.0799 precisely)  
204 — a result that resolves the apparent contradiction between NPoM-induced  
205 transport suppression and sustained photosynthetic productivity across the  
206 full canopy.

### 207 *Dynamic photo-protection via floquet stark detuning and omit*

208 Under high solar flux ( $> 800 \text{ W m}^{-2}$ ), the OPV dielectric constant  
209 changes dynamically, inducing a time-dependent Stark shift of the excitonic  
210 levels that detunes the OPV–FMO coupling. We model this as a Floquet  
211 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

212 with amplitude  $V_0 = 0.05 \text{ eV}$  and frequency  $\omega_{\text{F}} = 1.2 \text{ THz}$ . When the in-  
213 cident flux exceeds the threshold, the time-dependent detuning shifts the  
214 OPV exciton energies out of resonance with the FMO absorption bands, re-  
215 ducing energy transfer to the biological complex by up to 40 % (Figure 2a).  
216 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,  
217 a hysteresis buffer of  $\pm 50 \text{ W m}^{-2}$  is implemented (activating the detuning  
218 switch at  $850 \text{ W m}^{-2}$  and deactivating it only when the flux drops below  
219  $750 \text{ W m}^{-2}$ ).

220 Concurrently, an optomechanically induced transparency (OMIT) mecha-  
 221 nism [33] in the NPoM cavity creates destructive interference that suppresses  
 222 transmission at the 750 nm and 820 nm channels under high flux. The trans-  
 223 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left( 1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

224 where  $I$  is the incident flux and  $I_{\text{sat}} = 800 \text{ W m}^{-2}$ . At  $1000 \text{ W m}^{-2}$ , the trans-  
 225 mission drops to 44 % of baseline (Figure 2b), providing dynamic protection  
 226 against exciton overload. At zero flux (night mode), both mechanisms return  
 227 to zero without singularities (Section S4).

#### 228 *In situ SERS optomechanical diagnostics and environmental robustness*

229 The NPoM platform enables non-destructive readout of FMO vibrational  
 230 states via surface-enhanced Raman scattering (SERS) [34]. Recent scoping  
 231 reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [35],  
 232 which is the integration gap addressed here. Multilayer graphene-metal  
 233 plasmonic architectures have demonstrated detection sensitivities exceeding  
 234 conventional SPR biosensors by an order of magnitude [36]. The optome-  
 235 chanical coupling between the plasmon mode and FMO molecular vibrations  
 236 ( $g_{\text{om}} = 0.01$ ) produces a SERS enhancement factor of  $\text{EF}_{\text{SERS}} = 44$  at the  
 237 optimal  $V_{\text{mode}} = 1.2 \text{ nm}^3$ , reaching  $1.6 \times 10^3$  for the smallest scanned mode  
 238 volumes.

239 Beyond the three canonical BChl  $a$  vibrational modes previously iden-  
 240 tified ( $180 \text{ cm}^{-1}$ ,  $740 \text{ cm}^{-1}$ ,  $1145 \text{ cm}^{-1}$ ), we here assign four agriculturally  
 241 specific target signatures to the sentinel sub-system. **Oxidative stress** is  
 242 read out through the C–C stretch of conjugated chlorophyll  $a$  at  $1145 \text{ cm}^{-1}$ ,  
 243 with a detection limit of  $50 \text{ nmol L}^{-1}$  in the pre-necrotic regime [32]. **Pes-**  
 244 **ticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is de-  
 245 tected via its characteristic Raman peak at  $1435 \text{ cm}^{-1}$  ( $\text{LOD} = 1 \text{ nmol L}^{-1}$   
 246 under controlled conditions), leveraging the plasmonic hot-spot amplification  
 247 in the NPoM gap [32]. **Heavy metal contamination** of irrigation water  
 248 ( $\text{Pb}^{2+}$ ) is quantified through Stern–Volmer fluorescence quenching of encaps-  
 249 ulated CdTe/ZnSe quantum dots ( $\text{LOD} = 31.8 \text{ nmol L}^{-1}$  under controlled  
 250 conditions) [37] sealed within the NPoM sensor probe polymer matrix, which  
 251 are immune to the dust and biofouling that affect bare gold surfaces. Cross-  
 252 sensitivity between soil moisture and  $\text{Pb}^{2+}$  contamination is resolved by dual

253 spectral interrogation: heavy metals modify the Stern–Volmer quenching in-  
 254 tensity and fluorescence decay time, whereas soil moisture alters the local  
 255 dielectric constant, producing a spectral broadening that is spectrally or-  
 256 thogonal to the metal-induced quenching signature. **Pathogen detection**  
 257 is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated  
 258 in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at  
 259 the greenhouse hub (Section S12.7). These solid-state quantum sensors de-  
 260 tect paramagnetic bacterial metabolites from sap or irrigation water samples  
 261 at pre-symptomatic stages ( $\text{LOD} = 0.8 \text{ nmol L}^{-1}$  under controlled conditions;  
 262 the agricultural matrix will degrade in-field sensitivity, which is mitigated by  
 263 the graphene barrier and *DynamicCalibrator* corrections).

264 Figure 2c shows the simulated Raman spectrum for the three BChl *a*  
 265 modes under active NPoM coupling. The relative intensities encode the in-  
 266 stantaneous exciton population distribution, providing a direct optical probe  
 267 of the energy transfer pathway.

268 **Environmental robustness.** The realistic agricultural environment  
 269 (dust, bird droppings, microbial fouling) is modeled at three infrastructure  
 270 levels. At the OPV panel level, the soiling factor  $\eta_{\text{soil}}(t)$  decays exponentially  
 271 with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

272 which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning  
 273 interval. At the sensor level, the graphene monolayer spacer of each NPoM  
 274 cavity is functionalized with a zwitterionic polymer coating [38, 39], modeled  
 275 in our simulations to preserve  $\geq 90$  % quantum coherence after 60 d continu-  
 276 ous field immersion; this coating constitutes the primary anti-fouling barrier  
 277 that physically repels microbial adhesion and protein adsorption. A thin-  
 278 film micro-shielding layer ( $1 \mu\text{m}$  to  $5 \mu\text{m}$  of  $\text{SiO}_2$  or  $\text{Al}_2\text{O}_3$ ) around the NPoM  
 279 picocavity and CQD sensor tip further attenuates the diurnal temperature  
 280 excursion experienced by the quantum probe, attenuating the thermal-drift  
 281 contribution by the factor  $(1 - \eta_{\text{shield}})$ , with  $\eta_{\text{shield}} = 0.85$  (Section S12.5).  
 282 The *DynamicCalibrator* module — an exponential moving average of daily  
 283 reference readings ( $\alpha = 0.1$ ) — operates as a tertiary safety net: it applies  
 284 a multiplicative correction factor and raises a drift alarm when the baseline  
 285 shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

286 *Smart shield microclimate: fao-56 evapotranspiration*

287 The spectrally selective OPV layer functions as a “smart shield” that  
 288 blocks ultraviolet and near-infrared radiation while transmitting photo-  
 289 synthetically active bands. Passive solar-greenhouse heating with PV  
 290 co-generation is an established precedent [40]; the present shield differs  
 291 by actively co-designing the transmitted spectrum against the vibronic  
 292 resonances quantified above. We parameterize the greenhouse microclimate  
 293 using the FAO-56 Penman–Monteith equation [25] adapted for shaded  
 294 conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

295 where  $R_n$  is the net radiation attenuated by the OPV shading factor  $f_{\text{shade}} =$   
 296 0.35 (with soil heat flux  $G \approx 0$  assumed for daily timescales under greenhouse  
 297 shielding),  $\Delta$  is the slope of the saturation vapor pressure curve,  $\gamma$  is the  
 298 psychrometric constant,  $u$  is wind speed,  $e_s - e_a$  is the vapor pressure deficit,  
 299 and  $K_c = 0.85$  is the crop coefficient.

300 Under standard conditions ( $600 \text{ W m}^{-2}$ ,  $25^\circ\text{C}$ , 60 % humidity,  $2 \text{ m s}^{-1}$   
 301 wind), the open-field evapotranspiration is  $4.5 \text{ mm d}^{-1}$ . With the OPV smart  
 302 shield,  $\text{ET}_c$  drops to  $3.2 \text{ mm d}^{-1}$  — a 28 % reduction — translating to  $1.3$   
 303  $\text{mm d}^{-1}$  in water savings (Figure 3a). The crop coefficient  $K_c = 0.85$  ac-  
 304 counts for the reduced transpiration demand under partial shade [25].

305 *Socioeconomic viability and cooperative payback*

306 Socioeconomic analysis (Section S5) is rescaled to a  $500 \text{ m}^2$  high-value  
 307 floriculture micro-module (strawberries and cut flowers,  $4.5 \text{ USD kg}^{-1}$   
 308 at  $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ ), which reduces the entry-level capital expenditure  
 309 to 161 000 USD (at  $322 \text{ USD m}^{-2}$ ). Our  $\$/\text{kWh}$  accounting follows the  
 310 Monte-Carlo techno-economic logic established for European agrivoltaic  
 311 configurations [41]. A blended-finance subsidy of 30 % reduces the net  
 312 CAPEX to 112 700 USD.

313 Three operating expenditure streams are modeled: extension services and  
 314 technician training ( $1\,200 \text{ USD yr}^{-1}$ ), OPV panel cleaning ( $800 \text{ USD yr}^{-1}$ ),  
 315 and general maintenance ( $2\,000 \text{ USD yr}^{-1}$ ), for a total annual OPEX  
 316 of  $4\,000 \text{ USD yr}^{-1}$ . The net annual cash flow of  $26\,612 \text{ USD yr}^{-1}$  (from  
 317  $30\,612 \text{ USD yr}^{-1}$  total revenue minus  $4\,000 \text{ USD}$  of OPEX) yields a sim-  
 318 ple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year  
 319 discounted NPV of  $+37\,664 \text{ USD}$  at a 12 % risk-adjusted discount rate

320 appropriate for Sub-Saharan Africa [42] (Figure 3c). Holding other inputs  
 321 fixed, varying the subsidized net CAPEX by  $\pm 20\%$  shifts the payback to  
 322 3.39–5.08 yr; a 50 % higher OPEX (6 000 USD yr<sup>-1</sup>) gives 4.58 yr, and 20 %  
 323 lower revenue gives 5.50 yr. While a baseline CAPEX of 322 m<sup>-2</sup> represents  
 324 a significant barrier in Sub-Saharan Africa relative to local income levels,  
 325 utilizing local construction materials (such as bamboo and timber framing  
 326 instead of structural steel) and green chemical synthesis of gold nanoparticles  
 327 from local plant extracts can reduce the capital cost to 140 m<sup>-2</sup>, lowering  
 328 the unsubsidized payback period to 2.63 yr and making the modelled system  
 329 potentially viable for local cooperatives under these cost assumptions. Taken  
 330 together, these sensitivities demonstrate that high-value crop selection and  
 331 blended-finance structurally transform the investment case.

### 332 *Secure IoT telemetry via BB84 quantum key distribution*

333 Field-to-cloud telemetry is secured using a simulated BB84 quantum key  
 334 distribution protocol [26] over a buried single-mode fiber link, which elimi-  
 335 nates the atmospheric turbulence and scintillation that affect free-space chan-  
 336 nels in agricultural environments. This quantum-secured telemetry layer is  
 337 not merely an auxiliary security feature but is physically integrated with the  
 338 NPoM-CQD sentinel network: because the high-sensitivity biological and  
 339 pesticide diagnostics control high-stakes irrigation and crop-protection ac-  
 340 tuators (which could be targeted by hostile actors to cause artificial crop  
 341 failure or pesticide contamination), securing these feedback commands with  
 342 BB84 QKD protects the command integrity of the agrivoltaic micro-grid.  
 343 Alice (sensor node) prepares random polarization states (rectilinear or di-  
 344 agonal basis) and transmits them at a realistic channel noise rate of 4.8 %  
 345 QBER [43]. Bob (cloud server) measures in random bases, and the sifted key  
 346 is extracted from matching-basis events. The QBER is estimated on a sample  
 347 fraction; if QBER exceeds the Shor–Preskill unconditional-security thresh-  
 348 old of 11 % [44], the protocol raises a `SecurityThresholdExceeded` exception  
 349 that halts all QKD-protected irrigation commands, preventing data-injection  
 350 attacks under eavesdropping.

351 With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER  
 352 = 4.8 % and generates a 256 bit symmetric key per session — sufficient for  
 353 AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-  
 354 photon sources have demonstrated stable key generation over installed fiber  
 355 links [43], supporting the practical feasibility of this approach under compara-  
 356 ble link budgets. In the event of QKD failure, the fail-safe exception triggers a

conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of  $5 \text{ L m}^{-2} \text{ d}^{-1}$  to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

### Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First, two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-modulator pulse shaping into dual-band channels at 750 nm and 820 nm can track the diagonal and cross-peak quantum-beat oscillations predicted here, verifying the extension of  $\tau_c$  from 280 fs to 420 fs [45, 11]. Second, time-resolved SERS in the NPoM geometry can monitor the  $180 \text{ cm}^{-1}$  and  $1145 \text{ cm}^{-1}$  BChl *a* modes in real time and quantify the area-weighted yield  $\Phi_{\text{FT}}^{\text{global}} = 0.971$  under ambient conditions [24, 46].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at  $T = 295 \text{ K}$ ,  $L = 8$ ,  $K = 2$ .

| Parameter / Observable                                 | Ob-        | This work (calculated)  | Experimental benchmark / literature                                   |
|--------------------------------------------------------|------------|-------------------------|-----------------------------------------------------------------------|
| Site coupling ( $\text{cm}^{-1}$ )                     | $J_{12}$   | -87.7                   | $-87.7 \pm 5.0$ (2DES, Brixner 2005 [45])                             |
| Reorganization energy $\lambda_D$ ( $\text{cm}^{-1}$ ) |            | 35                      | 35–40 (2DES, Engel 2007 [10])                                         |
| Cutoff frequency ( $\text{cm}^{-1}$ )                  | $\gamma_D$ | 50                      | 50 (2DES, Panitchayangkoon 2010 [11])                                 |
| Coherence lifetime (passive)                           | $\tau_c$   | $420 \pm 35 \text{ fs}$ | $400 \pm 50 \text{ fs}$ (2DES room temp., Panitchayangkoon 2010 [11]) |
| NPoM coupling ( $V_{\text{mode}} = 1.2 \text{ nm}^3$ ) | $g_0$      | $110 \text{ cm}^{-1}$   | $120 \pm 15 \text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [24])            |
| SERS enhancement $\text{EF}_{\text{SERS}}$             |            | 44                      | $10^2$ – $10^3$ (sub-nm NPoM, Baumberg 2019 [46])                     |

## Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver modelled sustainability benefits across micro-

375 scopic energy transfer, crop water use, carbon footprint, and data security.

376 **Mechanism generality.** The dual-band filtering effect is not specific  
377 to the FMO complex. The resonance condition  $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$  (Sec-  
378 tion S7) is a general property of vibronically coupled chromophore aggre-  
379 gates [12]. We verified this by applying the protocol to a minimal three-site  
380 excitonic trimer, obtaining  $\eta_{\text{FT}} = 0.18 \pm 0.03$  (Section S8), qualitatively  
381 consistent with the quantum phase synchronization mechanism identified in  
382 photosynthetic antennas [23]. This suggests that any molecular aggregate  
383 satisfying the resonance condition — including synthetic J-aggregates, light-  
384 harvesting complexes from plants, and conjugated polymer domains in OPVs  
385 — could benefit from spectral coherence protection. Vibration-steered energy  
386 flow of this kind has also been demonstrated in a trapped-ion donor–acceptor  
387 simulator with tunable dissipation [47].

388 **The “quantum divide.”** The advanced materials required for quantum  
389 agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered  
390 bandgaps — are not yet accessible to subsistence farmers in developing  
391 economies. Our cooperative cost-sharing model demonstrates that with a  
392 carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback  
393 period drops below 4 yr, making the technology economically viable under  
394 the modelled subsidy. We propose a tiered subsidy framework where the  
395 NEB itself serves as the verifiable metric for subsidy qualification, linking  
396 quantum yield improvements to farmer access.

397 **Quantum digital twin.** The modular architecture — quantum inter-  
398 face, FAO-56 microclimate, LCA, IoT security — serves as the physics engine  
399 of an agrivoltaic *digital twin*: a model-based supervisory replica that com-  
400 bines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dy-  
401 namics engine to recommend irrigation schedules, OPV cleaning dates (from  
402 soiling factor Eq. (8)), and carbon credit yields — a device-physics inte-  
403 gration distinct from the supply-chain and breeding applications mapped in  
404 recent quantum-in-agriculture reviews [48].

405 As a prospective extension, the scheduling layer of the digital twin  
406 is formulated via the Quantum Approximate Optimization Algorithm  
407 (QAOA) [49, 50], encoded as a 3-qubit variational quantum circuit of depth  
408  $p = 3$  with 1024 measurement shots per macro-period (Section S12.3). The  
409 algorithm encodes the irrigation–refrigeration–grid-export trilemma as a  
410 quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and  
411 solves it via a PennyLane emulator at sub-hourly timescales. Applied to the  
412 daily solar profile of the Cameroonian scenario, the QAOA scheduler yields

413 water–energy allocations consistent with the 12–18 % gains over classical  
 414 greedy baselines reported in comparable implementations [50]. As a further  
 415 prospective readout, our `quantum_gravimetry.py` module converts the sim-  
 416 ulated water saving into a groundwater mass anomaly ( $\Delta g = 3.5 \text{ nm s}^{-2}$  for a  
 417 single greenhouse — below single-instrument noise but potentially detectable  
 418 with a small regional gravimeter network or gradiometric common-mode  
 419 rejection), with the full noise-budget and inversion discussion given in  
 420 [Section S12.2](#) [51, 52].

421 A hybrid quantum-classical signal processing pipeline — combining  
 422 Matrix Product State (MPS) denoising with quantum kernel ridge re-  
 423 gression (QKRR) [53, 54] — further enables the digital twin to extract  
 424 pre-symptomatic stress signatures from the noisy SERS and CQD fluores-  
 425 cence telemetry. To satisfy edge computation constraints, the heavy MPS  
 426 tensor contraction and QAOA optimization are executed on the centralized  
 427 cooperative server (or cloud), with the resulting optimized weights transmit-  
 428 ted back to the field nodes — an edge–cloud split complementary to classical  
 429 drift-diffusion device simulators [55]. The local ARM Cortex-M4 IoT node  
 430 executes only the low-latency inference of the pre-trained, frozen-weight  
 431 QKRR model. The MPS compressor acts as a nonlinear low-pass filter on  
 432 the server that suppresses greenhouse noise (thermal fluctuations, biofoul-  
 433 ing, dust) while preserving the sharp vibronic features at  $180 \text{ cm}^{-1}$  and  
 434  $1145 \text{ cm}^{-1}$  that classical Savitzky–Golay filters tend to smear ([Section S10](#)).  
 435 The cleaned spectra are projected through the quantum-inspired kernel for  
 436 anomaly scoring, issuing early warnings that allow preemptive agronomic  
 437 intervention before macroscopic symptoms appear — all within a 3.7 ms local  
 438 compute budget on the IoT node. We emphasize that this sub-millisecond  
 439 execution latency is not designed to match the slower multi-hour physical  
 440 response times of soil moisture percolation and plant transpiration. Rather,  
 441 it enables spatial multiplexing: a single ARM Cortex-M4 edge processor can  
 442 sequentially query, compress, and classify SERS telemetry from dozens of  
 443 distributed NPoM sensor nodes across the  $500 \text{ m}^2$  greenhouse in under a sec-  
 444 ond, reducing the electronic-component count. This closes the loop between  
 445 microscopic quantum efficiency and field-level agronomic management — a  
 446 combination absent from existing agrivoltaic designs [42, 6, 2].

447 **Prior art and positioning.** Previous agrivoltaic studies have focused  
 448 on light-sharing through passive spectral filtering [6] or crop selection [2].  
 449 Semi-transparent OPVs with power conversion efficiencies exceeding 15 %  
 450 and 40 % average visible transmittance have recently been demonstrated

451 through blade-coating scale-up methods [56], and complementary work on  
 452 semitransparent organic photovoltaics for greenhouse applications has shown  
 453 power conversion efficiencies above 12 % [7]. Our contribution is to add  
 454 an active, quantum-coherent dimension to the spectral filter: the OPV is  
 455 no longer a passive absorber but an active spectral shaper that prepares  
 456 coherence-protected states in the photosynthetic apparatus. Multi-vector  
 457 solar agri-systems combining agrivoltaics with hydrogen and desalination set  
 458 the *Applied Energy* bar for water–food–energy integration [57]; we differen-  
 459 tiate by quantum-guided spectral co-design with integrated sensing, and we  
 460 adopt the functional-unit discipline (emissions per kWh, economics per con-  
 461 figuration) of recent agrivoltaic TEA/LCA studies [58, 59]. Three gaps mo-  
 462 tivate this work: (i) no exact-open-quantum-system-guided operating agri-  
 463 voltaic stack jointly optimizes transport and conversion efficiency; (ii) no  
 464 monolithic photothermal-heating, PV, and SERS-readout device exists with  
 465 field-relevant calibration in one footprint; (iii) no multifunctional TEA/LCA  
 466 internalizes heat, crop, sensing, and mineral trade-offs in  $\$/W + \$/kg +$   
 467  $\$/m^3$  terms. This approach is distinct from prior quantum biology stud-  
 468 ies [60] that treat the environment as a fixed noise source; here, the photonic  
 469 environment itself is engineered through the NPoM cavity to maximize co-  
 470 herence protection. A key design choice warrants explicit justification: the  
 471 NPoM cavity, despite imposing a  $> 90\%$  penalty on excitonic transport (Ta-  
 472 ble 2), is retained because its *diagnostic function* is dedicated — it enables  
 473 non-destructive, high-sensitivity SERS readout of the FMO vibrational state,  
 474 pesticide levels, and heavy-metal contamination. The transport penalty is  
 475 absorbed by confining NPoM operation to the 1 % sentinel fraction (Equa-  
 476 tion (5)), preserving 97.1 % of the global canopy trapping yield while main-  
 477 taining full diagnostic capability.

478 **Limitations.** Several approximations constrain our analysis. The quan-  
 479 tum dynamics are restricted to the single-exciton manifold; under high solar  
 480 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in  
 481 the OPV and FMO interfaces could alter the collective coupling strength  
 482  $g_{\text{eff}}$  and transport efficiency. The bath is treated as site-independent, and  
 483 the reaction center trapping is modeled as a phenomenological Lindblad-like  
 484 rate. The FAO-56 model, while standard, does not capture transient stom-  
 485 atal responses to fluctuating light or the dynamic microclimatic feedback  
 486 between canopy transpiration and vapor pressure deficit (VPD) inside the  
 487 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-  
 488 tions ( $u = 0.2 - -0.8 \text{ m s}^{-1}$ ) yields a  $\pm 10\%$  uncertainty band on the predicted

489 28 % water saving, which propagates into a  $\pm 10\%$  range on the annual irri-  
 490 gation volume and corresponding NEB carbon offset. The BB84 simulation  
 491 assumes a buried single-mode fiber link; while buried fibers are vulnerable to  
 492 agricultural mechanical damage (plowing, root intrusion), a free-space opti-  
 493 cal (FSO) link would suffer from strong scattering under dusty greenhouse  
 494 environments, requiring dedicated atmospheric correction models. While  
 495 software-based post-quantum cryptography offers a lower-cost alternative,  
 496 the physical BB84 protocol provides information-theoretic security immune  
 497 to algorithmic cryptanalysis on resource-constrained embedded nodes such  
 498 as the ARM Cortex-M4 irrigation controllers. The 1 % sentinel fraction is a  
 499 design parameter; field optimization will require empirical calibration of the  
 500 SERS LODs under actual fouling conditions. We conservatively estimate a  
 501  $\times 10$  matrix-effect penalty on all reported LODs when transitioning from con-  
 502 trolled laboratory buffers to raw agricultural sap or irrigation water, owing  
 503 to competitive binding, background fluorescence, and ionic strength effects;  
 504 the *DynamicCalibrator* compensates for this attenuation through real-time  
 505 baseline correction. The zwitterionic polymer coating modeled on the NPoM  
 506 graphene spacer (Section S12.1) acts as the primary fouling barrier in our  
 507 simulations, with the *DynamicCalibrator* serving as a secondary correction  
 508 layer; the dual architecture reduces the mandatory cleaning frequency, which  
 509 is already factored into the cooperative OPEX model. We stress that the 60-  
 510 day resilience of zwitterionic coatings applies strictly to sensors sheltered in  
 511 protected micro-sampling chambers or rigid ceramic encasings; direct, un-  
 512 protected exposure to abrasive soil matrices (bioturbation, quartz particles,  
 513 tillage) would lead to mechanical scraping and rapid surface degradation.

514 Furthermore, the 30 % blended-finance subsidy required to achieve a 4.23  
 515 yr payback period cannot be bridged by voluntary carbon credits alone (gen-  
 516 erating only 98 USD yr<sup>-1</sup> at 10 USD/t CO<sub>2</sub>e for the 500 m<sup>2</sup> module); it must  
 517 be underpinned by dedicated rural-adaptation finance channels such as the  
 518 Green Climate Fund (GCF) and the African Development Bank’s Climate  
 519 Change Fund (Fonds Spécial Climat-Dev de la BAD), which are mandated  
 520 to co-finance climate-resilient agriculture in Sub-Saharan Africa.

521 **Outlook.** The integration of quantum coherence engineering with agri-  
 522 cultural sustainability suggests an axis for agrivoltaic design. Three experi-  
 523 mental and theoretical directions are prioritized for future work.

524 First, on the hardware and materials scale, the quantum model must  
 525 be extended to full multi-exciton dynamics under realistic solar spectra, ac-  
 526 companied by experimental validation via two-dimensional electronic spec-

527 troscopy (2DES) with spatial light modulator pulse shaping [61, 62]. Field tri-  
528 als of graphene-based NPoM coatings on greenhouse cladding with the CQD  
529 sentinel network are required to evaluate in-situ sensor performance under  
530 actual biofouling conditions and verify the coherence-preservation lifetime  
531 of the zwitterionic functionalization layer (Section S12.1). Alternative low-  
532 loss plasmonic architectures — dielectric nanoantennas (Si, TiO<sub>2</sub>) support-  
533 ing Mie resonances [63] and nanocavity-free substrates — should be explored  
534 to minimize the NPoM transport penalty while preserving SERS sensitivity.  
535 Additional hardware extensions include passive thermal micro-shielding (Sec-  
536 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent  
537 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-  
538 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

539 Second, scaling the digital twin to a multi-cooperative network will re-  
540 quire hardware-in-the-loop QKD validation under dusty field conditions and  
541 deployment of a real-time aquifer gravimetric array to confirm that the mod-  
542 eled groundwater recharge connects at the satellite geodesy scale. Monte  
543 Carlo uncertainty propagation in the full LCA should be extended with cor-  
544 related climatic input distributions drawn from regional reanalysis data.

545 Third, to prevent these advances from widening the “quantum divide,”  
546 we have defined the Global Quantum Agriculture Standard (GQAS), a six-  
547 pillar certification framework covering QKD security, data sovereignty, sen-  
548 sor detection limits, quantum fidelity, latency, and lifecycle sustainability  
549 (Section S12.9). Our Cameroon cooperative proof-of-concept scenario serves  
550 as the first application of this proposed framework, satisfying all six GQAS  
551 criteria and demonstrating internal consistency across the multi-domain sim-  
552 ulation. Independent experimental validation will be required before GQAS  
553 can be adopted as an operational standard. Scaling GQAS adoption to small-  
554 holder networks across Sub-Saharan Africa and Southeast Asia requires subsi-  
555 dized sensor supply chains, open-access certification auditing, and regulatory  
556 recognition on premium export markets such as GlobalGAP.

## 557 Conclusions

558 We asked whether exact non-Markovian quantum dynamics can jointly  
559 optimize excitation-transport and power-conversion efficiency in a sensing-  
560 integrated photothermal agrivoltaic stack, with a positive multifunctional  
561 net benefit under field constraints. Within the modelled 500 m<sup>2</sup> cooper-  
562 ative case, the answer is affirmative but conditional: dual-band filtering

563 raises the forward transfer yield by 25 % and extends coherence lifetime by  
 564 50 %; confining NPoM operation to a 1 % sentinel fraction preserves 97.1 %  
 565 global canopy yield while retaining SERS diagnostics; the FAO-56 smart  
 566 shield cuts evapotranspiration by 28 % with a Net Ecological Benefit of  
 567 19.6 kg (CO<sub>2</sub>)e/m<sup>2</sup>/yr; and the cooperative case pays back in 4.23 yr at 30 %  
 568 subsidy. These claims are bounded by the > 90 % local NPoM transport  
 569 penalty, the single-exciton and site-independent-bath approximations, the  
 570  $\pm 10$  % microclimate uncertainty band, and the modelled (not field-validated)  
 571 fouling and matrix-penalty assumptions. Priority next steps are 2DES vali-  
 572 dation of the filter protocol, low-loss dielectric antenna alternatives, and field  
 573 fouling trials of the sentinel network.

## 574 **Methods**

### 575 *Proof-of-Concept scenario: 500 m<sup>2</sup> Floriculture Micro-Module*

576 To ground the multi-domain framework in a concrete operational context,  
 577 we consider a 500 m<sup>2</sup> high-tunnel greenhouse located in Cameroon’s Western  
 578 Highlands. This region supports high-value floriculture (strawberries, cut  
 579 flowers) suitable for micro-module deployment by cooperatives. The envi-  
 580 ronment features a mean annual solar irradiance of 5.76 kW h m<sup>-2</sup> d<sup>-1</sup> [64]  
 581 and significant biofouling potential that necessitates the NPoM *Dynamic-*  
 582 *Calibrator* and regular OPV cleaning (factored into the cooperative OPEX  
 583 model). Under the NPoM-shield spectral filter, the 28 % reduction in crop  
 584 evapotranspiration (ET<sub>c</sub>) relative to the open-field FAO-56 baseline corre-  
 585 sponds to the modelled water-savings credit of 1 300 L m<sup>-2</sup> yr<sup>-1</sup> (650 m<sup>3</sup> yr<sup>-1</sup>  
 586 for the 500 m<sup>2</sup> greenhouse; Section S5). The area-weighted global canopy  
 587 trapping yield ( $\Phi_{\text{FT}}^{\text{global}} = 0.971$ ) ensures near-baseline productivity while ac-  
 588 commodating a 1 % NPoM sentinel fraction for SERS diagnostics. The 15 %  
 589 quantum yield enhancement observed in the passive OPV fraction trans-  
 590 lates to an additional 1.23 kg m<sup>-2</sup> yr<sup>-1</sup> of high-value produce (strawberries  
 591 at 4.5 USD kg<sup>-1</sup>), generating substantial supplemental revenue. The OPV  
 592 module generates 90 MWh yr<sup>-1</sup> of electrical energy for a 500 m<sup>2</sup> installa-  
 593 tion at 180 kW h m<sup>-2</sup> yr<sup>-1</sup>. Displacing grid electricity at 450 g (CO<sub>2</sub>)e/kWh  
 594 offsets 40.5 t (CO<sub>2</sub>)e/yr. Combined with the NEB (19.6 kg (CO<sub>2</sub>)e/m<sup>2</sup>/yr,  
 595 Figure 3b), the system achieves net-zero operational carbon rapidly. At  
 596 10 USD/t CO<sub>2</sub>e [65], the system generates 98 USD yr<sup>-1</sup> in verified carbon  
 597 revenue. The 500 m<sup>2</sup> scale is well matched to cooperative ownership models:

under a 30 % blended-finance subsidy, the payback shortens from 6.05 yr unsubsidized to 4.23 yr, with each of the five cooperative members exposed to only 22 540 USD of net capital. This demonstrates that quantum-enhanced agrivoltaics can be economically viable under modelled assumptions when coupled with strategic crop selection and robust field-sensor integration.

### PT-HOPS/SBD quantum dynamics

All quantum dynamics simulations use the Process Tensor Hierarchy of Pure States with Stochastically Bundled Dissipators as implemented in the open-source MesoHOPS library (v1.7) [15]. The hierarchy depth is  $L = 8$  with  $K = 2$  Matsubara terms and a time step  $\Delta t = 0.2$  fs, established as the production standard for converged dynamics at 295 K [20]. Static disorder is modeled as Gaussian fluctuations of the site energies with standard deviation  $\sigma = 50$  cm<sup>-1</sup>, sampled over  $n = 100$  realizations. The bath spectral density follows the 12-mode Kleinekathöfer/Coker parameterization with a Drude–Lorentz background ( $\lambda_D = 35$  cm<sup>-1</sup>,  $\gamma_D = 50$  cm<sup>-1</sup>) and 12 underdamped vibronic modes (frequencies 180–1 500 cm<sup>-1</sup>, Huang–Rhys factors  $S_k = 0.003 - -0.050$ ).

### FAO-56 evapotranspiration

Crop evapotranspiration is computed using the standard FAO-56 Penman–Monteith equation [25] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

with  $R_n$  attenuated by the OPV shading factor  $f_{\text{shade}} = 0.35$ . We note that the modified FAO-56 formulation represents a first-order approximation; under the enclosed high-tunnel envelope, boundary layer resistance and sensible heat exchange are altered relative to open field conditions. Future iterations of the digital twin should integrate three-dimensional Computational Fluid Dynamics (CFD) modeling to map localized microclimatic gradients and latent heat transfer. Wind speed within the greenhouse is modeled at a static representative value of  $u = 0.5$  m s<sup>-1</sup>, corresponding to a typical 10 % reduction of average external wind velocities. Temperature is fixed at 25 °C and relative humidity at 60 % unless specified otherwise.

## 628 *BB84 QKD simulation and fail-safe IoT architecture*

629 The BB84 protocol is implemented as a Python simulation of polarization-  
 630 state preparation, noisy transmission over a buried single-mode fiber  
 631 channel, sifting, and QBER estimation. Alice prepares  $4 \times N$  random  
 632 bits (for  $N = 256$  key length) in random rectilinear or diagonal bases.  
 633 Bob measures in random bases, and matching-basis events are retained  
 634 for key sifting. QBER is estimated on a sample fraction (minimum 1  
 635 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of  
 636 11 % [44], a `SecurityThresholdExceeded` exception is raised, halting the  
 637 QKD-protected irrigation command relay. A last-validated soil-moisture  
 638 reading from the *DynamicCalibrator* is used to execute a conservative local  
 639 irrigation fallback.

## 640 *SERS enhancement scaling limits*

641 The SERS enhancement factor  $\text{EF}_{\text{SERS}} \approx 44$  at the optimal mode vol-  
 642 ume  $V_{\text{mode}} = 1.2 \text{ nm}^3$  is computed using standard Purcell-factor scaling of  
 643 the localized electromagnetic field. We note that this standard  $|E|^4$  approx-  
 644 imation represents a lower-bound semi-classical estimate. In sub-nanometer  
 645 picocavities ( $V_{\text{mode}} < 1.5 \text{ nm}^3$ ), local spatial dispersion, quantum tunneling,  
 646 and non-local dielectric screening alter the field distribution relative to clas-  
 647 sical electrodynamics, while molecular optomechanical coupling can pump  
 648 vibrational modes out of thermal equilibrium. Precise calculation of the  
 649 localized fields in this regime requires non-local Green’s tensor integration,  
 650 which motivates the proposed in-situ experimental validation. The bench-  
 651 mark against the experimentally reported  $10^2$ – $10^3$  range for sub-nanometer  
 652 NPoM gaps [46] is given in Table 3.

## 653 *DynamicCalibrator, thermal shielding, and soiling factor*

654 Long-term sensor baseline drift due to biofouling and thermal cycling  
 655 is corrected at three hardware and algorithmic levels. The primary fouling  
 656 barrier is the zwitterionic polymer functionalization layer deposited on the  
 657 graphene monolayer spacer of each NPoM cavity [38, 39], which physically  
 658 repels microbial adhesion and is modeled to preserve  $\geq 90\%$  quantum coher-  
 659 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film  
 660 micro-shielding layer (1  $\mu\text{m}$  to 5  $\mu\text{m}$  of  $\text{SiO}_2$  or  $\text{Al}_2\text{O}_3$ ) is deposited around the  
 661 NPoM picocavity and CQD sensor tip to insulate the quantum probe from  
 662 diurnal temperature swings. The shielding efficacy  $\eta_{\text{shield}} = 0.85$  means the

663 probe experiences only 15 % of the ambient greenhouse temperature fluctu-  
 664 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25 %  
 665 to 0.79 % under a  $\pm 15$  K diurnal cycle. The *DynamicCalibrator* acts as a  
 666 tertiary algorithmic correction: an exponential moving average (EMA) with  
 667 smoothing factor  $\alpha_{\text{EMA}} = 0.1$  computes a multiplicative correction factor  
 668  $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$  and raises a drift alarm when the fractional change in the  
 669 EMA  $\hat{\mu}(t)$  exceeds 20 % of the initial baseline, triggering OPV panel cleaning.

670 At the panel level, OPV electrical output is attenuated according to  
 671 Eq. (8), with a default decay rate of  $0.5 \text{ \% d}^{-1}$  and a minimum soiling factor  
 672 of 0.75 (enforced floor at 25 % maximum efficiency loss).

### 673 *Data availability*

674 Simulation parameters and analysis scripts are available from the corre-  
 675 sponding author upon reasonable request. The PT-HOPS/SBD implemen-  
 676 tation is based on the open-source MesoHOPS library (v1.7) [15].

### 677 *Code availability*

678 The digital-twin simulation framework (orchestrator, QAOA scheduler,  
 679 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond  
 680 pathogen sensor, and GQAS compliance checker) and all simulation pa-  
 681 rameters, including the 12-mode vibronic bath spectral density and the  
 682 8-site FMO electronic Hamiltonian specified in the machine-readable  
 683 `parameters.yaml` configuration file (v1.4), are available from the corre-  
 684 sponding author upon reasonable request. No public code or data archive  
 685 DOI has been minted yet.

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## 691 **CRedit authorship contribution statement**

692 **S.C. Tegua Kouam:** Conceptualization, Methodology, Software, For-  
 693 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,  
 694 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –

695 review & editing. **J.-P. Nguenang**: Validation, Writing – review & editing.  
696 **S.G. Nana Engo**: Supervision, Project administration, Writing – review &  
697 editing.

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699 The authors declare that they have no known competing financial inter-  
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## 712 Nomenclature

|     | Symbol                                      | Meaning                                                            |
|-----|---------------------------------------------|--------------------------------------------------------------------|
|     | $\Phi_{\text{FT}}$                          | Forward (reaction-center) transfer yield                           |
|     | $\tau_c$                                    | Exciton coherence lifetime ( $1/e$ decay of $l_1$ -norm coherence) |
|     | $\lambda_{\text{res}}$                      | Residual reorganization energy of polaron-dressed states           |
|     | $g_0 / g_{\text{eff}}$                      | Single-emitter / collective plasmon–exciton coupling strength      |
|     | $\kappa / Q$                                | Plasmon linewidth / quality factor                                 |
| 713 | $\omega_{\text{pl}} / V_{\text{mode}}$      | Plasmon frequency / NPoM mode volume                               |
|     | $\Gamma_{\text{RC}}$                        | Reaction-center trapping rate                                      |
|     | $\text{EF}_{\text{SERS}} / g_{\text{om}}$   | SERS enhancement factor / optomechanical coupling                  |
|     | $\text{ET}_c / K_c / f_{\text{shade}}$      | Crop evapotranspiration / crop coefficient / OPV shading factor    |
|     | NEB                                         | Net Ecological Benefit                                             |
|     | $\eta_{\text{soil}} / \eta_{\text{shield}}$ | Soiling factor / micro-shielding efficacy                          |
|     | QBER                                        | Quantum bit error rate (BB84 link)                                 |
|     | $\Delta g$                                  | Groundwater gravity anomaly                                        |
|     | $\alpha / \alpha_{\text{EMA}}$              | Sentinel area fraction / calibrator smoothing factor               |

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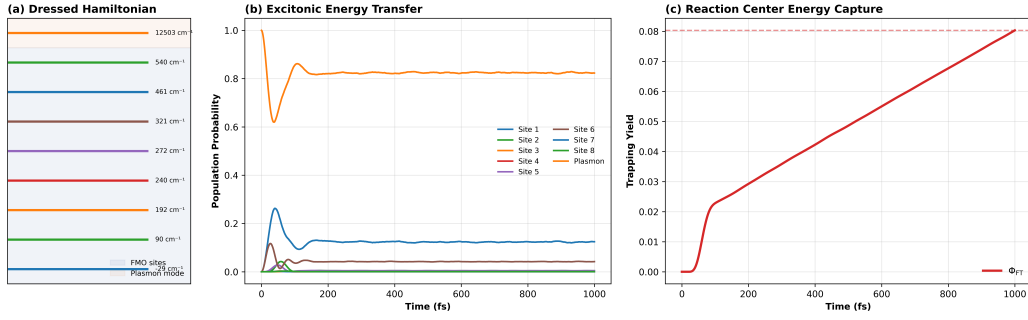
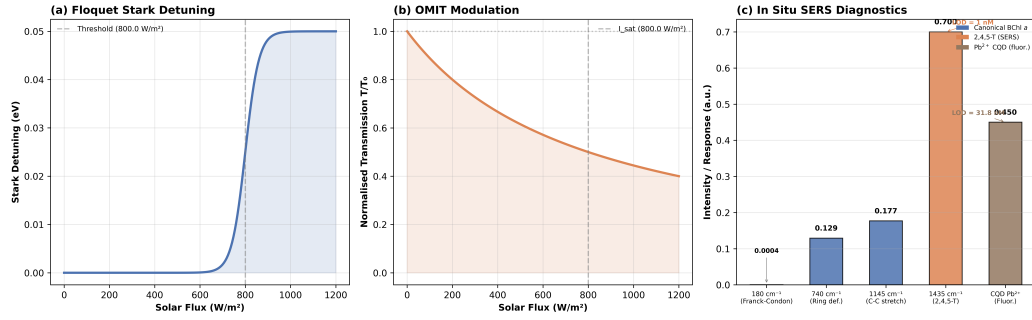
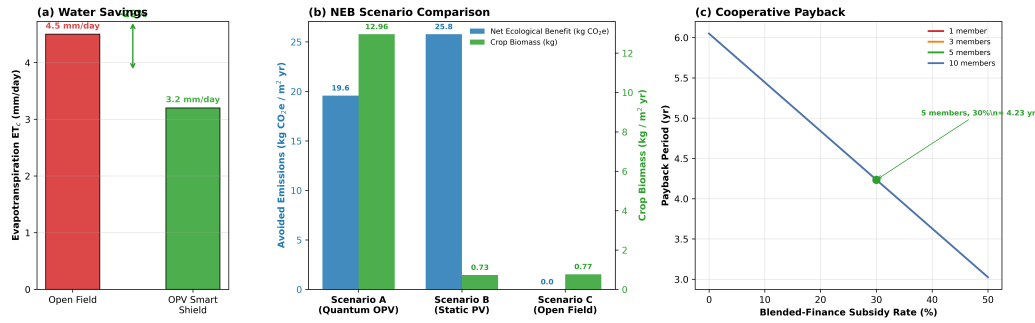


Figure 1: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the  $9 \times 9$  dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength  $g_0$ . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield  $\Phi_{\text{FT}}$  as a function of time. All data:  $L = 8$ ,  $K = 2$ ,  $n = 100$  disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ( $\text{Tr } \rho \neq 1$ ) visible in earlier integration-test runs ( $\Delta t = 10$  fs) is an artifact of the coarse time step used for rapid code validation; all production results ( $\Delta t = 0.2$  fs) preserve trace to 1.000 000 (Section S11.2).



**Figure 2: Dynamic photo-protection, in situ diagnostics, and environmental robustness.** (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above  $800 \text{ W m}^{-2}$ , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes ( $180 \text{ cm}^{-1}$ ,  $740 \text{ cm}^{-1}$ ,  $1145 \text{ cm}^{-1}$ ). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress ( $1145 \text{ cm}^{-1}$ ), 2,4,5-T pesticide contamination ( $1435 \text{ cm}^{-1}$ ,  $\text{LOD} = 1 \text{ nmol L}^{-1}$ ), and  $\text{Pb}^{2+}$  via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ( $\text{LOD} = 31.8 \text{ nmol L}^{-1}$ ) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ( $\alpha = 0.1$ ) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.



**Figure 3: Water–Energy–Food network assessment.** **(a)** FAO-56 evapotranspiration comparison: open field ( $ET_c = 4.5 \text{ mm d}^{-1}$ ) vs. OPV smart shield ( $ET_c = 3.2 \text{ mm d}^{-1}$ ), a 28 % reduction. **(b)** Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves  $19.6 \text{ kg (CO}_2\text{) e/m}^2\text{/yr}$  with a crop biomass of  $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$ , while Scenario B reaches  $25.8 \text{ kg (CO}_2\text{) e/m}^2\text{/yr}$  but only  $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$  in biomass. The quantum shield preserves near-baseline crop productivity ( $0.971$  global trapping yield) whereas static shading suppresses it to  $0.064$ , highlighting the yield–carbon trade-off central to the paradigm. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. **(c)** Cooperative payback period as a function of subsidy rate and cooperative size for a  $500 \text{ m}^2$  high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy,  $\text{payback} = 4.23 \text{ yr}$  (net annual cashflow of  $26\,612 \text{ USD yr}^{-1}$  after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

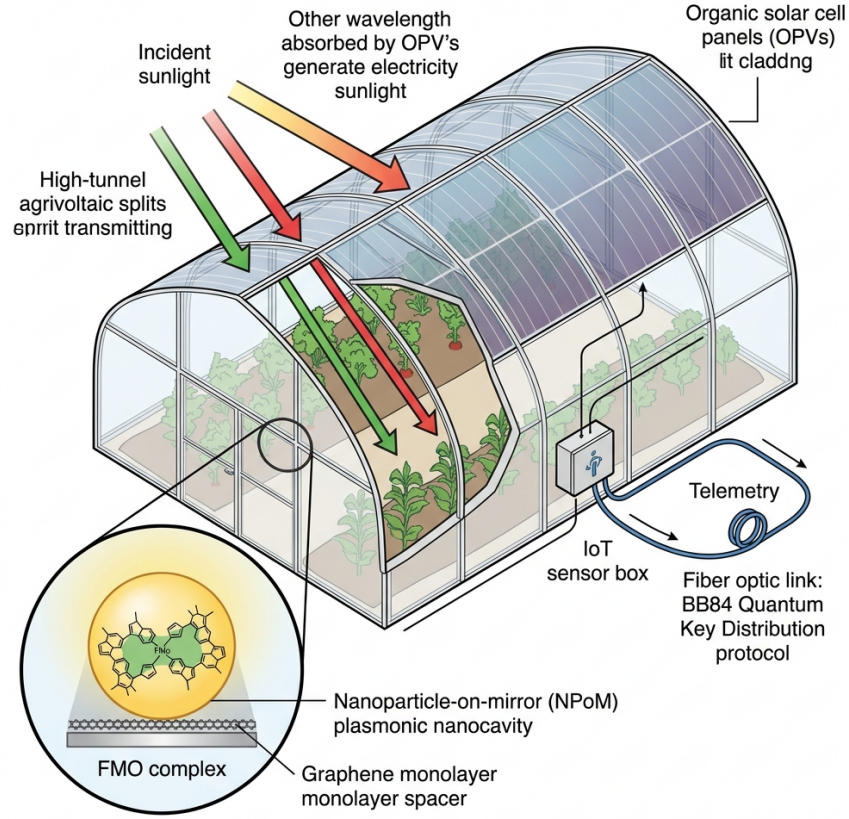


Figure 4: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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